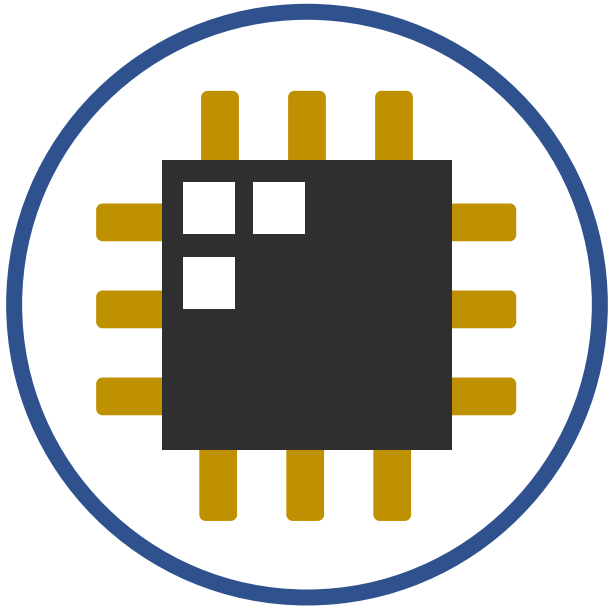
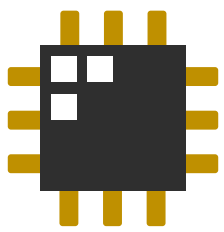




EEE 113:

Introduction to Electrical and Electronics Engineering Systems

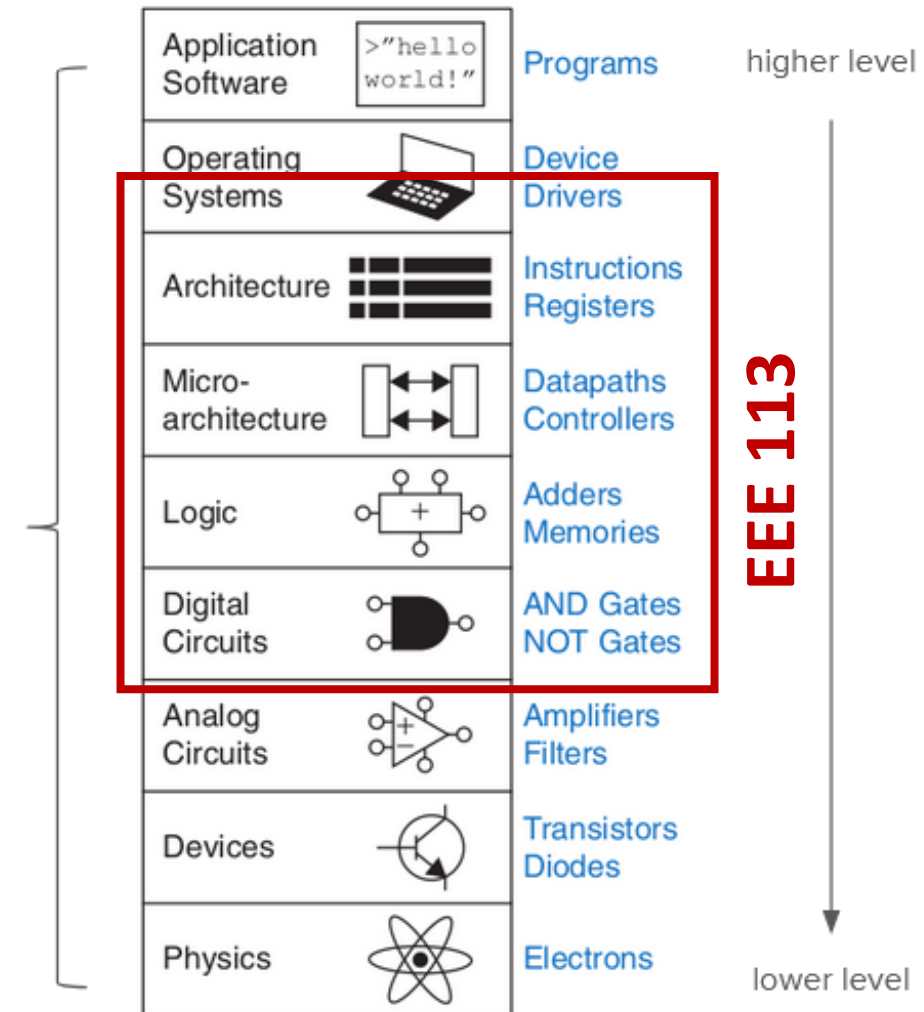
Week 1 –Number systems

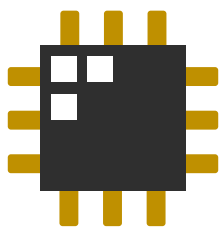


Introduction

- There are several abstraction layers in a computer system
 - EEE 111 is on the highest level (high-level programming using Python, C, etc)
- EEE 113 (this course) briefly touches the lower levels of hierarchy
 - How a computer *really* does what it does.
- Looking forward, we will be *very briefly* touching the topics covered in EEE 143 and EEE 153...
in 5 weeks!

Abstraction
Levels/Layers





Introduction

- For this iteration of EEE 113, we will be going from the bottom up



*Ready for the next
adventure?*

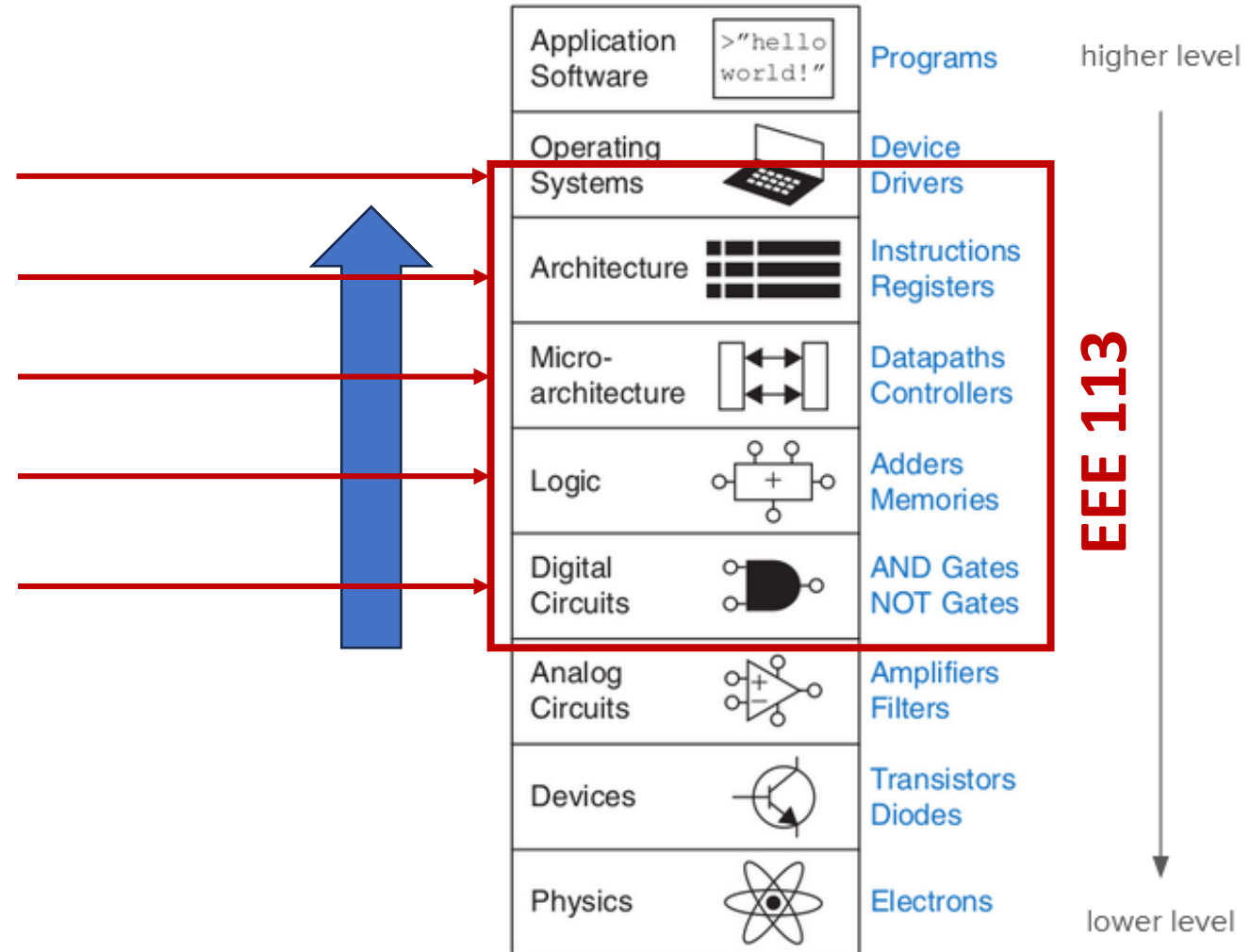
Week 5

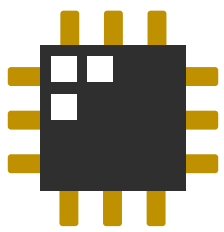
Week 4

Week 3

Week 2

Week 1



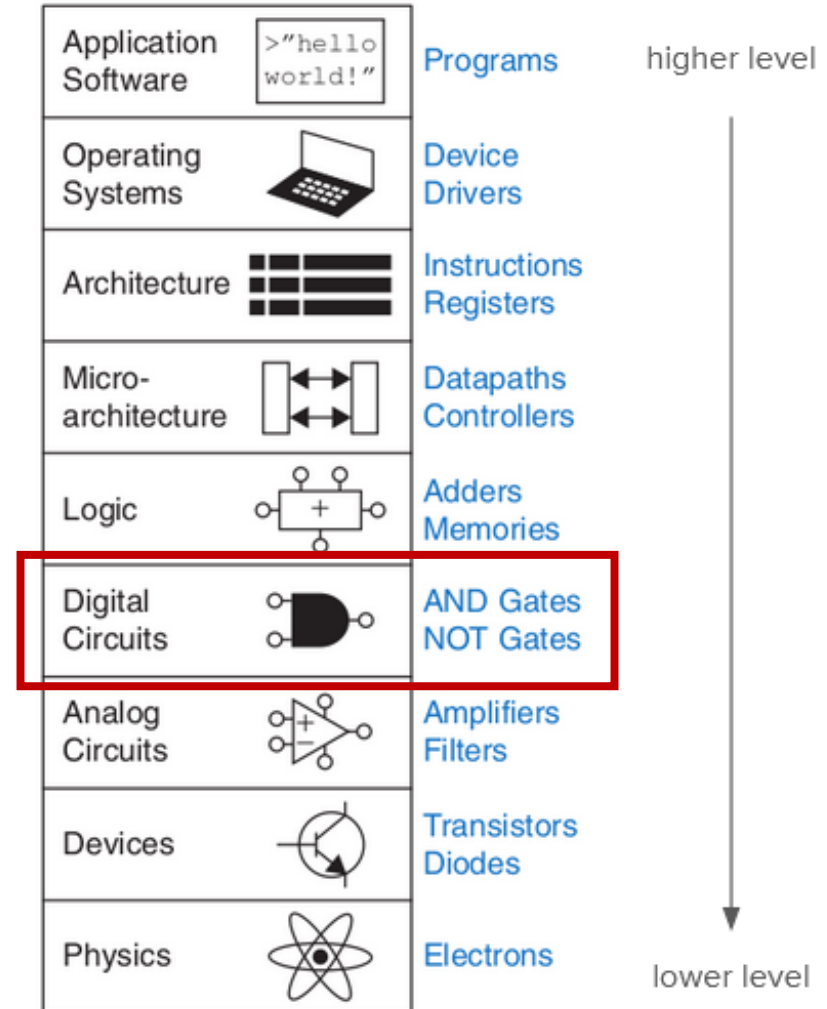


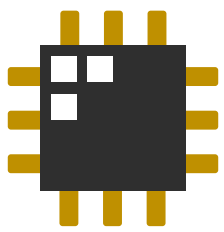
Introduction

- In Computing, we usually manipulate numbers in binary (base 2).
 - We only work in 0s and 1s!
 - We need to be comfortable with them



Week 1





Decimal numbers

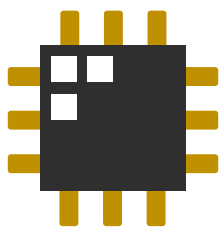
- We represent our numbers with 10 symbols or digits

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

- But counting does not end in 9...
- We would need more than 1 symbol to represent larger quantities

113

$$(\mathbf{1} \times 10^2) + (\mathbf{1} \times 10^1) + (\mathbf{3} \times 10^0)$$



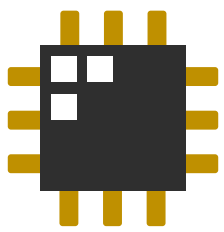
Decimal numbers

113

$$(1 \times 10^2) + (1 \times 10^1) + (3 \times 10^0)$$

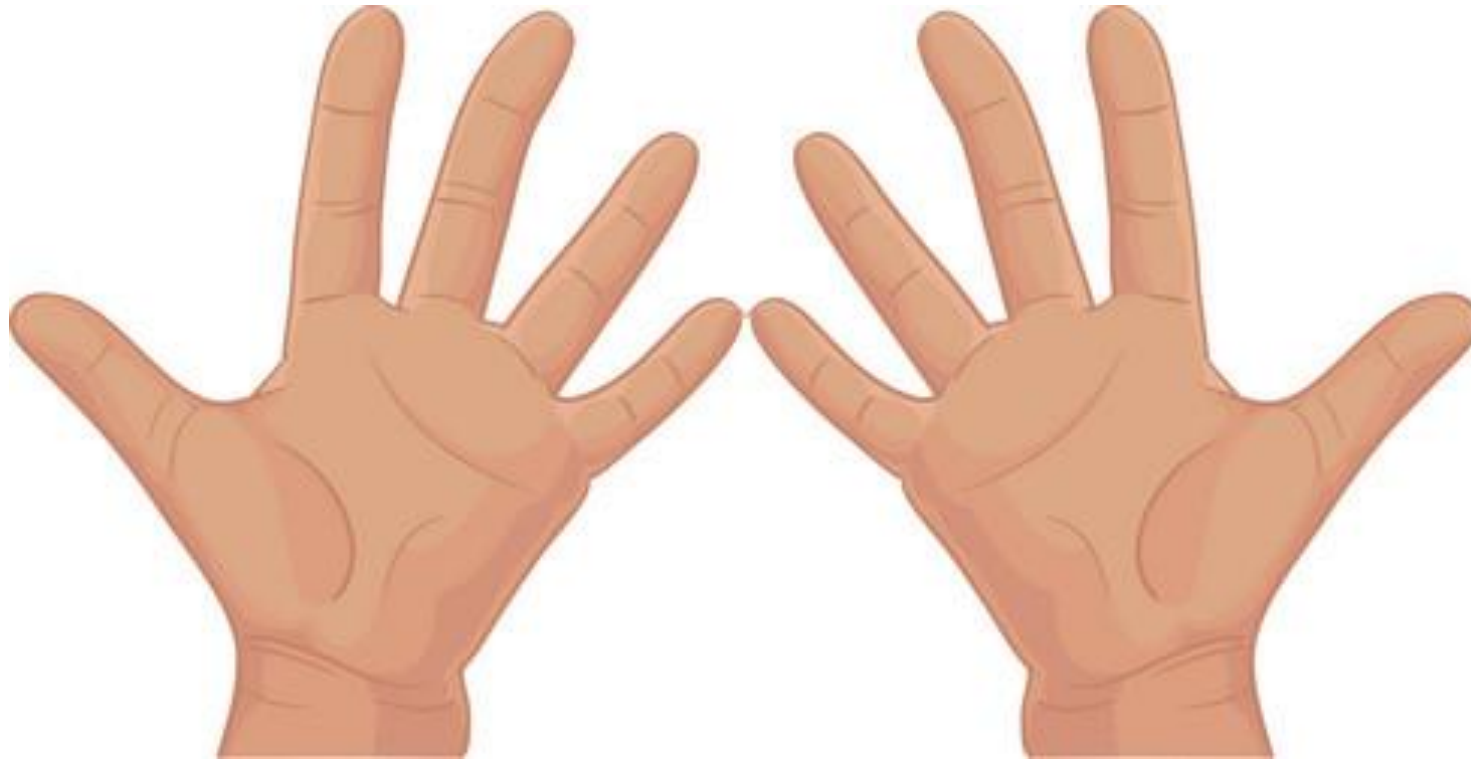
- Each digit position has different weighted values
- For decimal numbers, the digit on the left is 10 times higher than the digit on the right.

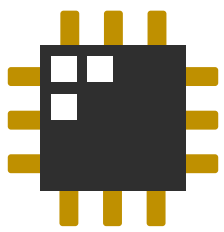
Decimal (from Latin *decimus*, meaning tenth)



Decimal numbers

- We have 10 fingers, so 10 comes out naturally!





Other number systems

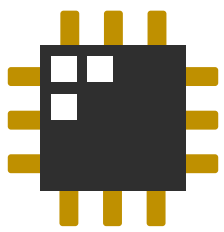
113

$$(1 \times 10^2) + (1 \times 10^1) + (3 \times 10^0)$$

- We have 10 digits, every digit position is a power of 10...

- What if we have less?





Binary numbers

- We can represent any number in any base n , that has n digits with positional powers of n
- If we only have 2 digits, 0 and 1, how do we represent larger numbers?

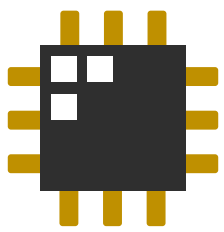
1110₂ (we say 1110 in base 2, or 1110 in binary)

$$(\textcolor{blue}{1} \times 2^3) + (\textcolor{orange}{1} \times 2^2) + (\textcolor{green}{1} \times 2^1) + (\textcolor{red}{0} \times 2^0)$$

2 digits (0,1) and positional powers of 2

$$1 \times 8 + 1 \times 4 + 1 \times 2 + 0 \times 1$$

$$= \textbf{14}_{10} \quad (\text{we say 14 in base 10, or 14 in decimal})$$



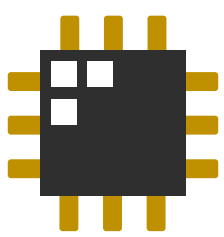
Binary numbers

- Base 2 is also called binary, we call the digits as *bits*

4-bit binary	Decimal value
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7

4-bit binary	Decimal value
1000	8
1001	9
1010	10
1011	11
1100	12
1101	13
1110	14
1111	15

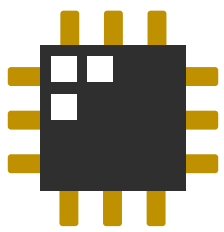
- If we need larger numbers, we need more bits!



Other bases

- Segue: How about base-1?



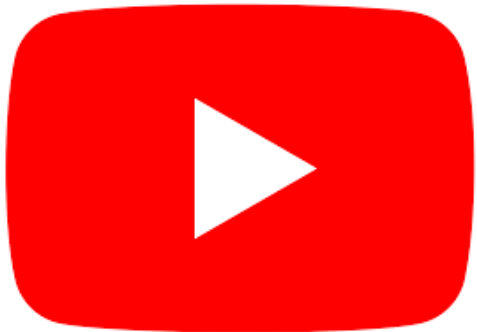


Other bases

- Segue: How about base-64?

<https://www.youtube.com/watch?v=dQw4w9WgXcQ>

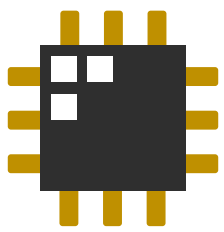
Trivia: Youtube **video IDs** are encoded in base-64!



Base 64 = 64 symbols

- a, b, c, ..., z = **26** symbols
- A, B, C, ..., Z = **26** symbols
- 0, 1, 2, ..., 9 = **10** symbols
- - and _ = **2** symbols

$$26+26+10+2 = 64$$



Binary number conversion

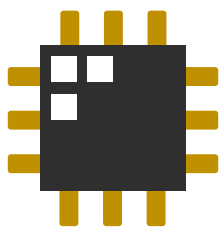
- This demonstrates conversion from binary to decimal
 - For every bit position **going to the left, we increase the power of 2**, then just **multiply the actual bit** to that power of 2

1110₂

$$(\mathbf{1} \times 2^{\mathbf{3}}) + (\mathbf{1} \times 2^{\mathbf{2}}) + (\mathbf{1} \times 2^{\mathbf{1}}) + (\mathbf{0} \times 2^{\mathbf{0}})$$

$$\mathbf{1} \times 8 \quad + \quad \mathbf{1} \times 4 \quad + \quad \mathbf{1} \times 2 \quad + \quad \mathbf{0} \times 1$$

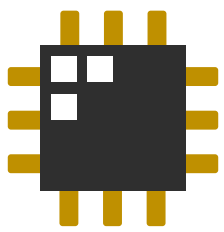
$$= \mathbf{14}_{10}$$



Binary number conversion

- Practice, what is the decimal equivalent of this?

1110001_2



Binary number conversion

- Practice, what is the decimal equivalent of this?

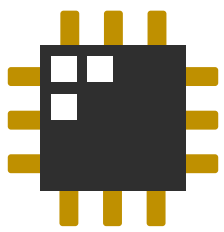
1110001_2

$$(\underline{1} \times 2^6) + (\underline{1} \times 2^5) + (\underline{1} \times 2^4) + (\underline{0} \times 2^3) + (\underline{0} \times 2^2) + (\underline{0} \times 2^1) + (\underline{1} \times 2^0)$$

$$(\underline{1} \times 64) + (\underline{1} \times 32) + (\underline{1} \times 16) + (\underline{0} \times 8) + (\underline{0} \times 4) + (\underline{0} \times 2) + (\underline{1} \times 1)$$

$$64 + 32 + 16 + 0 + 0 + 0 + 1$$

113_{10}



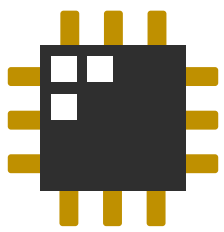
Binary number conversion

- How to do the reverse? Brute-force way: Successive division
 - Divide the number by the base (in this case 2). Get the quotient.
 - Continuously divide the quotient by the base until it becomes 0.
 - Always take note of the remainder each time the quotient is divided.
 - Arrange the order of remainders. The last remainder is at the left-most position.

$$\begin{array}{l} 14 / 2 = 7 \text{ remainder } 0 \\ 7 / 2 = 3 \text{ remainder } 1 \\ 3 / 2 = 1 \text{ remainder } 1 \\ 1 / 2 = 0 \text{ remainder } 1 \end{array}$$


Read in this direction

1110₂

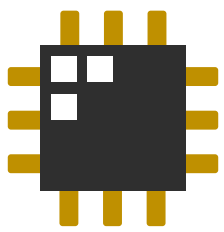


Binary number conversion

- How to do the reverse? Faster(?) way
 - Memorize the powers of 2.
 - Subtract from the number the highest power of 2 that is less than that number.
 - Work from the highest position (power of 2), then work towards the right continually subtracting the power of 2. Don't subtract if it becomes negative.

We are basically doing the process in reverse (starting from the highest power of 2 possible)
Highest power of 2 less than 14 is $8 = 2^3$. We start from there.

$$(_x2^3) + (_x2^2) + (_x2^1) + (_x2^0) = 14$$

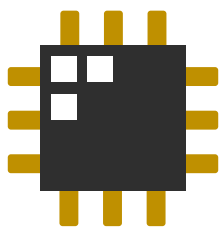


Binary number conversion

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$$\begin{array}{ccccccc} (1 \times 2^3) & + & (_ \times 2^2) & + & (_ \times 2^1) & + & (_ \times 2^0) = 14 \\ 8 & + & ? & + & ? & + & ? = 14 \end{array}$$

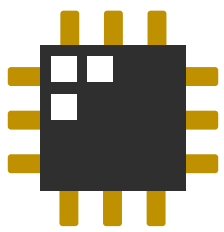


Binary number conversion

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Highest power of 2 less than 14 is $8 = 2^3$. We start from there.

$$\begin{array}{ccccccc} (1 \times 2^3) & + & (1 \times 2^2) & + & (_ \times 2^1) & + & (_ \times 2^0) = 14 \\ 8 & + & 4 & + & ? & + & ? = 14 \end{array}$$

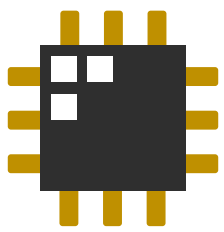


Binary number conversion

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 - Subtract from the number the highest power of 2 that is less than that number.
 - Work from the highest position (power of 2), then work towards the right continually subtracting the power of 2. Don't subtract if it becomes negative.

We are basically doing the process in reverse (starting from the highest power of 2 possible)
Highest power of 2 less than 14 is $8 = 2^3$. We start from there.

$$\begin{aligned} (1 \times 2^3) + (1 \times 2^2) + (1 \times 2^1) + (_ \times 2^0) &= 14 \\ 8 + 4 + 2 + ? &= 14 \end{aligned}$$

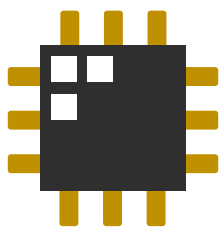


Binary number conversion

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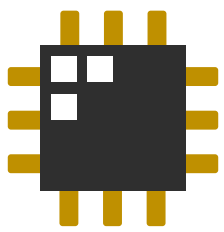
$$\begin{aligned} &(\mathbf{1} \times 2^3) + (\mathbf{1} \times 2^2) + (\mathbf{1} \times 2^1) + (\mathbf{0} \times 2^0) = 14 \\ &8 \quad + \quad 4 \quad + \quad 2 \quad + \quad 0 \quad = 14 \end{aligned}$$



Binary number conversion

- Practice, write 43 in binary. You can use either method.

43_{10}



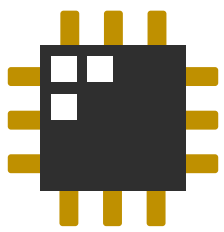
Hexadecimal numbers

- Another common base is base 16 (called hexadecimal or hex)

4-bit binary	Hex value	Decimal value
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7

4-bit binary	Hex value	Decimal value
1000	8	8
1001	9	9
1010	A	10
1011	B	11
1100	C	12
1101	D	13
1110	E	14
1111	F	15

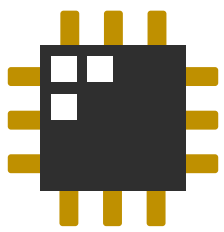
- In hex, there are 16 digits (0 to F).
- We usually write it with 0x prefix (e.g. 0xA, 0xF, 0xAF)
- It might be helpful to memorize this table as they come up a lot!



Binary to hex conversion

- Practice, write 1010 1101 1111 0001 in hex

1010 1101 1110 0001₂



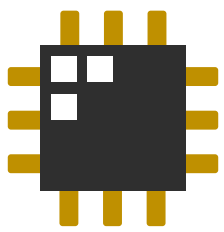
Binary to hex conversion

- Practice, write 1010 1101 1111 0001 in hex

1010 1101 1110 0001₂

A D E 1

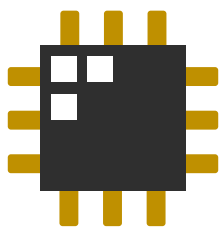
0xADE1 or ADE1₁₆



Hex to decimal conversion

- Practice, what is the decimal equivalent of 0xAF?

$$\mathbf{0xAF = AF_{16}}$$



Hex to decimal conversion

- Practice, what is the decimal equivalent of 0xAF?

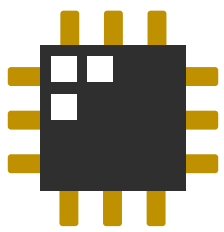
$$\mathbf{0xAF = AF_{16}}$$

$$(\mathbf{\underline{A}} \times 16^1) + (\mathbf{\underline{F}} \times 16^0) = ?$$

$$(\mathbf{\underline{10}} \times 16^1) + (\mathbf{\underline{15}} \times 16^0) = ?$$

$$(\mathbf{\underline{10}} \times 16) + (\mathbf{\underline{15}} \times 1) = ?$$

$$160 + 15 = \mathbf{175_{10}}$$

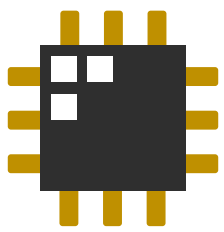


Hex to decimal conversion

- Practice, what is the decimal equivalent of 0xAF?

$$0x\textcolor{blue}{A}\textcolor{red}{F} = \textcolor{black}{AF}_{16}$$

$$\textcolor{blue}{1010} \textcolor{red}{1111}_2$$



Hex to decimal conversion

- Practice, what is the decimal equivalent of 0xAF?

$$0xAF = AF_{16}$$

$$1010\ 1111_2$$

$$(1 \times 2^7) + (0 \times 2^6) + (1 \times 2^5) + (0 \times 2^4) + (1 \times 2^3) + (1 \times 2^2) + (1 \times 2^1) + (1 \times 2^0)$$

$$(1 \times 128) + (0 \times 64) + (1 \times 32) + (0 \times 16) + (1 \times 8) + (1 \times 4) + (1 \times 2) + (1 \times 1)$$

$$128 + 0 + 32 + 0 + 8 + 4 + 2 + 1 = 175_{10}$$